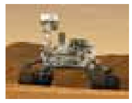
New African titanosaur
A new species discovered in Egypt's Western Desert will probably solve an old mystery looming around near the end of the Cretaceous-era. **http://dgit.in/MansouraD**



Mars Curiosity journey
NASA's Curiosity rover shot a panorama of the red planet that also showed its journey so far. **http://dgit.in/MarsCrsty**

CORAL REEFS

What do you know about the rainforests of the oceans?

Arnab Mukherjee | arnab@digit.in

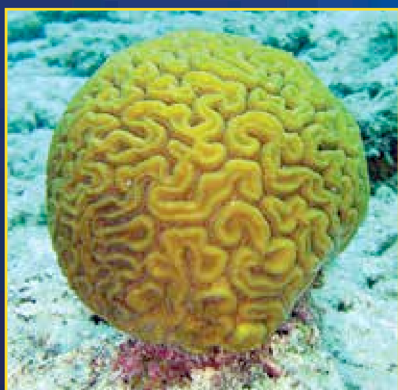
In any underwater documentary that you might have seen on channels like Discovery or National Geographic, amidst all the diverse flora and fauna, one particularly colourful and varying type of structure might have caught your attention. Yes, we are talking about coral reefs, the rainforests of the ocean. Coral reefs are underwater structures built of calcium carbonate that is secreted by corals. Corals are marine invertebrate animals that exist in colonies, and certain corals, who are designated reef builders, secrete the calcium carbonate that goes on to form the hardened exoskeleton that we now know as coral reefs. What makes this underwater structure unique is its diversity. While occupying merely 0.1% of the world's ocean's surface, they are home to more than 25% of its species and contribute in a multitude of ways to the environmental and geological stability of our planet. And at this point of time, they are in the worst condition they've ever been in. To understand why they're important what led to their current predicament, we have to start from the basics of coral reefs.

THE BUILDERS OF THE RAINFORESTS

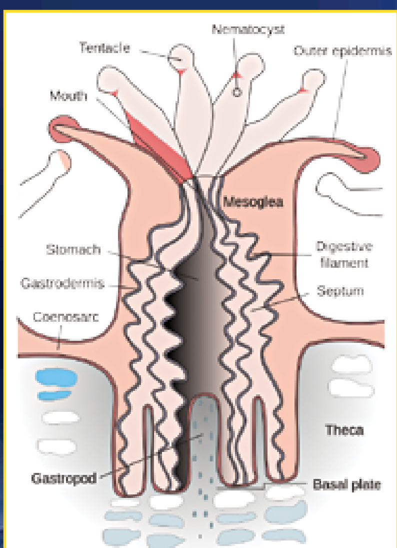
One common misconceptions associated with coral reefs is that they're alive. It would be akin to saying that hair, or nails, are alive by themselves. As stated earlier, coral reefs are formed by corals that are from the same family of creatures as sea anemones. That family is known as Cnidaria, and they have a common characteristic – each instance of an organism in this family is known as a polyp, whether it be a sea anemone or a coral. A polyp, in the case of corals, can be described as a tin can open at one end with tentacles coming out of said end. Coral reefs aren't built over the lifetime of a single coral – it takes hundreds of generations to produce the complex coral reef structures we see.

There are more than a hundred species of corals that can be found around the planet. Their exoskeleton colonies can form quite a visual display, from round, folded brain corals that resemble a human brain to tall, elegant sea whips and sea fans that look like intricate, vibrantly coloured trees or plants.

Corals, like almost all living things, survive by consuming other beings. They use their stinging tentacles to catch small sea life like plankton and small fish. Most corals, however, have a symbiotic relationship with algae called zooxanthellae (pronounced zo-zan-thel-ee). The algae that live inside the coral bodies are also a source of food. They make food for themselves and the coral, and in return get shelter and carbon dioxide. They're also the reason for all the bright colours, because most polyps are clear and colourless.



Brain coral (Diploria labyrinthiformis)



The anatomy of a coral polyp

REEFS

If all the coral reefs in the world were to be put together, they'd cover all of UK and Switzerland!